Sahaj Bhadja

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Summary

Full-stack and GenAI engineer with 2+ years building **Python**, **React**, and TypeScript systems. Experienced in RAG, multimodal LLMs, evaluation pipelines, human-in-the-loop workflows, prompt orchestration, observability, and dependable healthcare-adjacent applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and maintained a multi-tenant **React** and Python AI platform for 500+ users, optimizing human-review workflows, indexed evidence retrieval, and backend paths to reduce latency from 1.2 seconds to 350 milliseconds.
- Built monitored LLM workflows processing 5,000+ monthly jobs with structured outputs, validation, failure isolation, human review, and recovery paths, sustaining greater than 99% reliability across CI/CD-backed production releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python services handling 10,000+ daily requests, integrating GPT-4o, Claude 3.7, Qwen-VL, OCR, prompt workflows, and structured grading outputs with stable peak performance.
- Developed human-in-the-loop evaluation and concurrent processing for 1,000+ files with rubric validation, progress feedback, failure recovery, and cost tracking, reducing feedback turnaround 60% while maintaining 90%+ consistency.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **model-evaluation pipeline** processing 100,000+ records, improving planning quality 18% through reproducible experiments, automated metrics, controlled configurations, performance tracing, and structured analysis of failure modes.
- Implemented comparative evaluation across complex scenarios, improving closed-loop stability 15% while inspecting model outputs, debugging failure patterns, validating robustness, and documenting tradeoffs between learned behavior and expected outcomes.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered enterprise integration software with automated builds and VM-based validation across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito.
- Developed object-oriented microservices and AWS data pipelines using Glue, S3, SQL, caching, and pagination, improving response times 30%, processing efficiency 68%, and maintainability across integrated platforms with reliable operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, TypeScript, and Node.js features across eight microservices, integrating WebSockets and REST APIs to synchronize more than two million records across responsive customer and operational workflows.
- Automated testing and cloud delivery with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving release validation and production observability across environments.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
GenAI & Agents: LLM APIs, RAG, prompt chaining, retrieval pipelines, structured outputs, tool-use concepts, agentic workflows, LLaMA-2, GPT-4o, Claude, Qwen-VL
AI & Evaluation: PyTorch, Transformers, FAISS, multimodal AI, OCR, embeddings, semantic search, automated metrics, human-in-the-loop review, failure analysis, model monitoring
Full Stack: Python, React.js, Next.js, TypeScript, Flask, FastAPI, Node.js, REST APIs, PostgreSQL, MySQL, MongoDB, Azure SQL, WebSockets
Delivery & Tools: Git, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, AWS, Azure, GCP, Sentry, logging, observability, Claude Code, Codex, Cursor
Engineering Practices: Agile, rapid prototyping, unit testing, integration testing, debugging, code reviews, production support, cross-functional collaboration, technical documentation

Projects

Big-Brain | Live | *RAG, LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a RAG platform processing 50,000+ records with LLaMA-2, FAISS, evidence retrieval, summarization, secure access, and incremental indexing, reducing retrieval latency 40% through optimized embeddings and backend pipelines.

HMIS | Live | *React, TypeScript, REST APIs, Sentry*
- Built a role-based web application with scheduling, dashboards, notifications, validation, and Sentry monitoring, reducing administrative workload 30% while supporting secure access, maintainable components, and reliable post-launch operations.

PiXMod | Live | *Next.js, PyTorch, Transformers, MongoDB*
- Built a multimodal AI application using Next.js, Transformers, and MongoDB for segmentation, retrieval, and generative editing, processing 1,000+ assets through responsive human-review workflows, metadata storage, and cloud deployment.